More engineering varieties
Original web CAD + FEM references

Collected 12 October 2026 | Five source collections | No new CAD or FEM solutions generated

Engineering family	Available original reference
Hip implant contact and fixation	IGES components + Abaqus deck + numerical stress/pressure CSV
Aircraft wing vibration	IGES wing surfaces + NASTRAN decks + ten nodal mode-shape fields
Turbine blade dovetail dynamics	STEP assembly + published reduced mass and stiffness matrices
Aircraft fuselage crashworthiness	STEP shell CAD + Abaqus decks + numerical acceleration/DRI plots
Wire-arc additive manufacturing	STEP nominal wall/baseplate + VMAP thermal/distortion fields

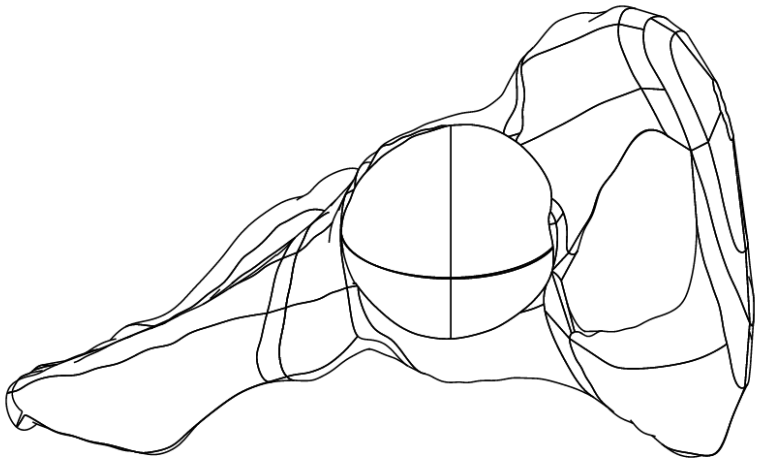
The reference types are different. Hip scalars and wing modal fields are solved numerical references. Blade matrices are FE operators; crash results are published plots requiring digitization for automatic scoring.

Five problem families. Four hip wear states, ten wing modes, sixteen blade preloads and three fuselage frame-count variants do not constitute 33 independent engineering task families. WAAM adds a fifth family with thermal and distortion subtask types.

Benchmark status: zero new certified admissions. Source CAD views follow; these are inspection renders of downloaded geometry, not verified dimensioned engineering drawings. The CAD-input track gains candidates; the drawing-input track remains incomplete.

All original assets, checksums, sources, reference types and limitations are in the accompanying dataset package. COMSOL sources are excluded. No model trials or solver runs were performed.

Hip implant contact and cement fixation



Inspection view of original pelvic IGES surfaces. Six original IGES files include pelvis, cement mantle and four cup wear states. Cups are solids; pelvis and cement import as valid surfaces.

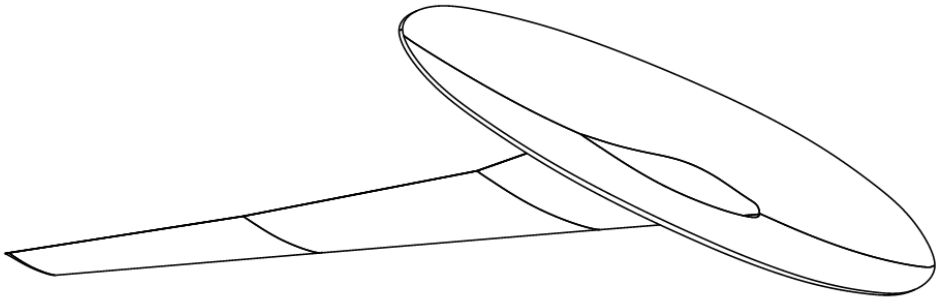
Wear mm	Cup VM MPa	Bearing p MPa	Cement VM MPa	Cement max principal MPa	Cement shear MPa
0.0	11.93	17.1	8.49	4.14	1.18
1.0	9.917	11.1	6.46	3.66	1.1
2.0	9.766	10.99	6.99	3.87	1.13
4.0	9.551	10.99	8.5	4.4	1.2

Published values copied from Results.csv for the CAD-matched 40 mm cup at 45 degrees. Other cup sizes/angles in the CSV lack matching CAD here. Input deck applies a 2500 N load with contact and nonlinear geometry; wear-state/deck mapping remains to verify.

No separate native head CAD was verified; it remains represented in the solver deck. Numerical replay and tolerances are not certified.

Source: Hua and Jin, University of Leeds. DOI 10.5518/13. CC BY 4.0.
archive.researchdata.leeds.ac.uk/15/

HIRENASD aircraft wing vibration



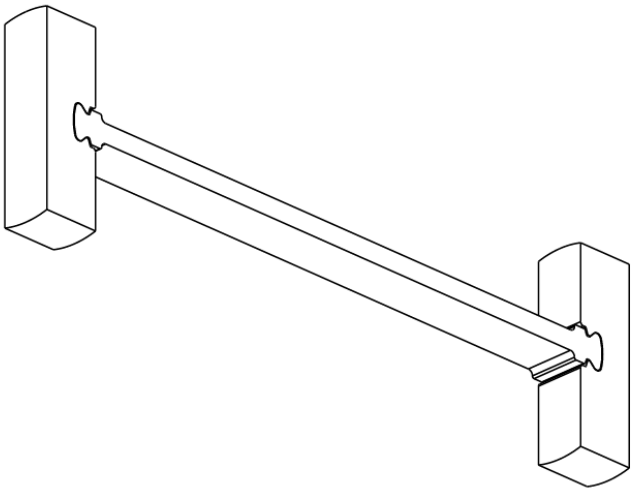
Inspection view of original IGES surface CAD: 14 valid faces. NASTRAN BDF archives and the original coordinate-system PDF are included.

Published content	Verified extent
Mode-shape fields	10 modes; 159,841 nodes per mode; all nodal values finite
Columns	X, Y, Z, node ID, f1, f2, f3
Units / geometry	CAD in mm; field coordinates in m. Extents agree after conversion.
Coordinate excerpt, mode 1	First node coordinates: (0.7818463135, 0.8, 0.0003451690376) m. Complete original rows included as an excerpt.

Fields are structural mode shapes interpolated onto a Centaur grid, not static displacements or stress fields. Sign, normalization, node mapping and frame conventions must be audited before mode-shape scoring. Frequency references are not included; the separate resource currently reports private access.

Source: NASA DASHlink HIRENASD / Aeroelastic Prediction Workshop distribution.
c3.ndc.nasa.gov/dashlink/resources/354/
Public download; dataset-specific license was not stated on inspected pages.

Turbine blade dovetail contact dynamics



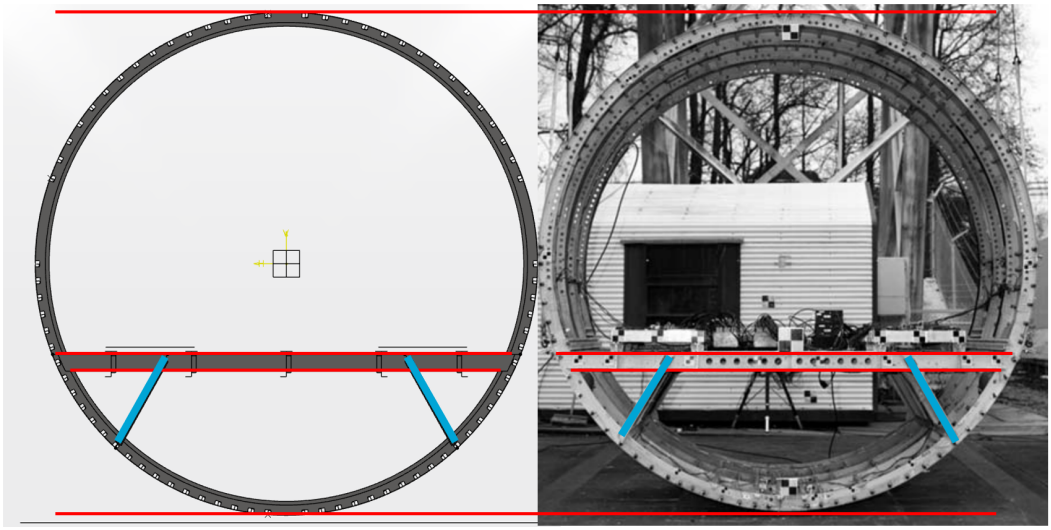
Inspection view of original mock-blade and two-slot STEP assembly: three valid solids, 102 faces. Separate blade and slot STEP/IGES files are included.

Published FEM object	Contents
M_GLOB	1968 x 1968 reduced mass matrix; finite and symmetric
K_GLOB_SE	1968 x 1968 x 16 reduced stiffness array; finite
CONNECTIVITY	132 x 6 contact-connectivity array
Preloads F_ax	1 N, then 2, 4, ... 30 kN
Source units	Mass: tons; stiffness: N/mm; positions: mm; load: N (Structures.xlsx).

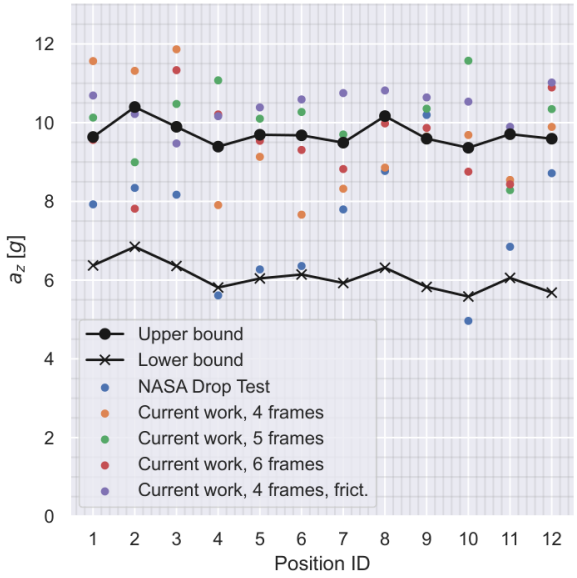
These are published FE reduced operators, not solved stress/displacement fields. Useful for reduction/operator tasks; they do not yet supply a full response ground truth. Experimental free-decay, frequency and damping data are separate and are not labeled as FEM results.

Source: Botto and Glorioso, Politecnico di Torino. DOI 10.17632/xkssd7gr92.1. CC BY 4.0. data.mendeley.com/datasets/xkssd7gr92/1

Fokker F28 fuselage crashworthiness



Original publisher image comparing CAD and physical fuselage. Downloaded STEP surface models have 978, 1142 and 1306 valid faces for 4-, 5- and 6-frame sections. Each has its original no-friction/no-damage Abaqus input deck.

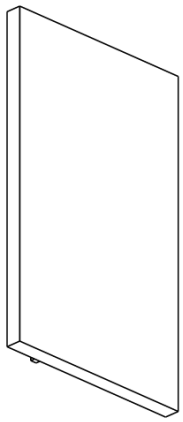


Original published mean acceleration plot: positions 1-12, vertical acceleration in g, with FEM curves and NASA experimental comparisons. Other original PDFs contain mesh-sensitivity acceleration and dynamic-response-index plots.

No raw CSV, ODB, full-field or time-history references verified in this subset. Plot digitization must distinguish FEM curves from experiments and carry uncertainty before numerical scoring.

Source: Desiderio et al., TU Delft. DOI 10.5281/zenodo.7724227. CC BY 4.0. zenodo.org/records/7724227

Wire-arc additive manufacturing



Inspection view of original WALLSpec.STEP: one valid solid, 11 faces. Published wall/baseplate model, material certificates/datasheets and manufacturing-procedure spreadsheet included.

Original result file	Published contents
M1 thermal, 30 s dwell	6000 states x 14,994 element temperatures; raw range 300-1654.91.
M1 distortion, 30 s dwell	115 states x 3885 nodes: displacement, temperature, stress, total/mechanical strain, force and PE.
M2 thermal, 30 s dwell	5000 states x 12,256 element temperatures; raw range 300-1502.00. Held geometry variant.

All selected result arrays contain finite values. Source archive members were downloaded directly and checked against their ZIP CRC. No local solver was run.

Geometry/unit review: M1 thermal extents agree with nominal CAD after axis permutation within 0.16 mm; exact registration is pending. M1 distortion coordinates match CAD in mm, contradicting a declared SI system. Raw stress/displacement units remain unverified. M2 model is about 5 mm wider and 7.4 mm shorter than nominal CAD; it is not counted as an exact CAD/result pair.

These are three result files from one manufacturing family, not 11,115 benchmark problems. No original executable solver deck or verified dimensioned drawing in this subset; setup/tolerance review remains pending.

Source: AIMEN, Fraunhofer SCAI and EBBAM, PIONEER project. DOI 10.5281/zenodo.17608626. CC BY 4.0. zenodo.org/records/17608626